

MC1488, SN55188, SN75188 QUADRUPLE LINE DRIVERS

SLLS094C – SEPTEMBER 1983 – REVISED MAY 2004

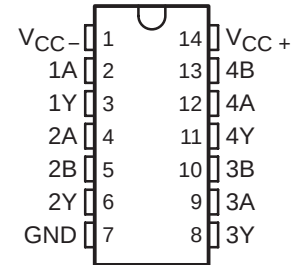
- Meet or Exceed the Requirements of ANSI TIA/EIA-232-E and ITU Recommendation V.28
- Current-Limited Output: 10 mA Typical
- Power-Off Output Impedance: 300 Ω Minimum
- Slew Rate Control by Load Capacitor
- Flexible Supply-Voltage Range
- Input Compatible With Most TTL Circuits

description/ordering information

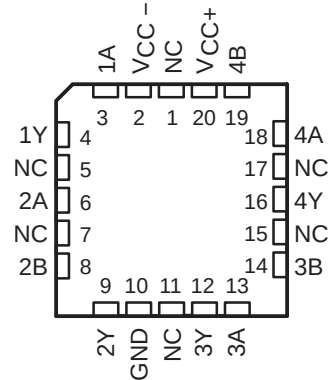
The MC1488, SN55188, and SN75188 are monolithic quadruple line drivers designed to interface data terminal equipment with data communications equipment in conformance with ANSI TIA/EIA-232-E, using a diode in series with each supply-voltage terminal as shown under typical applications.

The SN55188 is characterized for operation over the full military temperature range of -55°C to 125°C . The MC1488 and SN75188 are characterized for operation from 0°C to 70°C .

SN55188 . . . J OR W PACKAGE
SN75188 . . . D, N, OR NS PACKAGE
MC1488 . . . N PACKAGE
(TOP VIEW)



SN55188 . . . FK PACKAGE
(TOP VIEW)



NC – No internal connection

ORDERING INFORMATION

T_A	PACKAGE [†]		ORDERABLE PART NUMBER	TOP-SIDE MARKING
0°C to 70°C	PDIP (N)	Tube of 25	MC1488N	MC1488N
		Tube of 25	SN75188N	SN75188N
	SOIC (D)	Tube of 50	SN75188D	SN75188
		Reel of 2500	SN75188DR	
	SOP (NS)	Reel of 2000	SN75188NSR	SN75188
-55°C to 125°C	CDIP (J)	Tube of 25	SN55188J	SN55188J
			SNJ55188J	SNJ55188J
	CFP (W)	Tube of 150	SNJ55188W	SNJ55188W
	LCCC (FK)	Tube of 55	SNJ55188FK	SNJ55188FK

[†] Package drawings, standard packing quantities, thermal data, symbolization, and PCB design guidelines are available at www.ti.com/sc/package.



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PRODUCTION DATA information is current as of publication date. Products conform to specifications per the terms of Texas Instruments standard warranty. Production processing does not necessarily include testing of all parameters.

**TEXAS
INSTRUMENTS**

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On products compliant to MIL-PRF-38535, all parameters are tested unless otherwise noted. On all other products, production processing does not necessarily include testing of all parameters.

MC1488, SN55188, SN75188
QUADRUPLE LINE DRIVERS

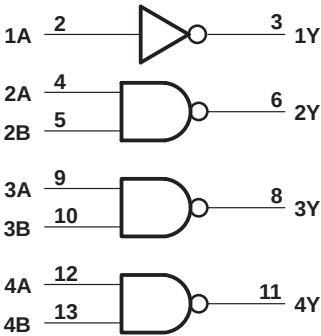
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FUNCTION TABLE
(drivers 2–4)

A	B	Y
H	H	L
L	X	H
X	L	H

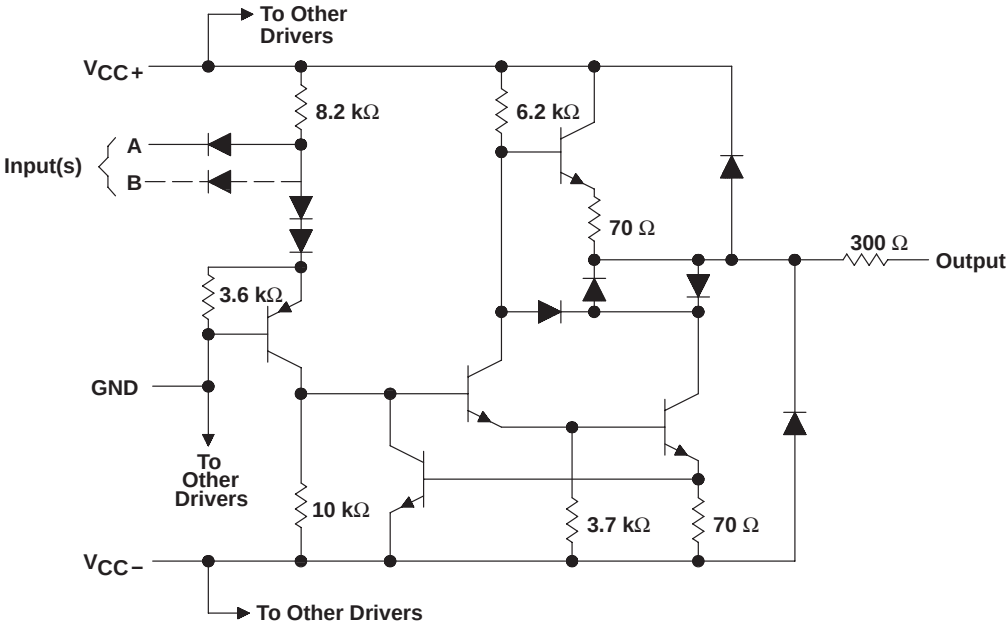
H = high level, L = low level,
X = irrelevant

logic diagram (positive logic)



Positive logic
 $Y = \overline{A}$ (driver 1)
 $Y = AB$ or $\overline{A} + \overline{B}$ (drivers 2 thru 4)

schematic (each driver)



Resistor values shown are nominal.

absolute maximum ratings over operating free-air temperature (unless otherwise noted)[†]

Supply voltage, V_{CC+} at (or below) 25°C free-air temperature (see Notes 1 and 2)	15 V
Supply voltage, V_{CC-} at (or below) 25°C free-air temperature (see Notes 1 and 2)	–15 V
Input voltage, V_I	–15 V to 7 V
Output voltage, V_O	–15 V to 15 V
Continuous total power dissipation (see Note 2)	See Dissipation Rating Table
Package thermal impedance, θ_{JA} (see Notes 3 and 4): D package	86°C/W
N package	80°C/W
NS package	76°C/W
Operating virtual junction temperature, T_J	150°C
Case temperature for 60 seconds, FK package	260°C
Lead temperature 1,6 mm (1/16 inch) from case for 60 seconds: J or W package	300°C
Storage temperature range, T_{Stg}	–65°C to 150°C

[†] Stresses beyond those listed under “absolute maximum ratings” may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under “recommended operating conditions” is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

- NOTES:
1. All voltage values are with respect to the network ground terminal.
 2. For operation above 25°C free-air temperature, refer to the maximum supply voltage curve, Figure 6. In the J package, SN55188 chips are alloy mounted.
 3. Maximum power dissipation is a function of $T_J(\max)$, θ_{JA} , and T_A . The maximum allowable power dissipation at any allowable ambient temperature is $P_D = (T_J(\max) - T_A)/\theta_{JA}$. Selecting the maximum of 150°C can affect reliability.
 4. The package thermal impedance is calculated in accordance with JESD 51-7.

DISSIPATION RATING TABLE

PACKAGE	$T_A \leq 25^\circ\text{C}$ POWER RATING	DERATING FACTOR ABOVE $T_A = 25^\circ\text{C}$	$T_A = 70^\circ\text{C}$ POWER RATING	$T_A = 125^\circ\text{C}$ POWER RATING
FK	1375 mW	11.0 mW/°C	880 mW	275 mW
J	1375 mW	11.0 mW/°C	880 mW	275 mW
W	1000 mW	8.0 mW/°C	640 mW	200 mW

recommended operating conditions

		SN55188			MC1488, SN75188			UNIT
		MIN	NOM	MAX	MIN	NOM	MAX	
V_{CC+}	Supply voltage	7.5	9	15	7.5	9	15	V
V_{CC-}	Supply voltage	–7.5	–9	–15	–7.5	–9	–15	V
V_{IH}	High-level input voltage	1.9			1.9			V
V_{IL}	Low-level input voltage			0.8			0.8	V
T_A	Operating free-air temperature	–55		125	0		70	°C

MC1488, SN55188, SN75188 QUADRUPLE LINE DRIVERS

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electrical characteristics over operating free-air temperature range, $V_{CC\pm} = \pm 9\text{ V}$ (unless otherwise noted)

PARAMETER		TEST CONDITIONS		SN55188			MC1488, SN75188			UNIT
				MIN	TYP†	MAX	MIN	TYP†	MAX	
V _{OH}	High-level output voltage	V _{IL} = 0.8 V, R _L = 3 kΩ	V _{CC+} = 9 V, V _{CC-} = -9 V	6	7		6	7		V
			V _{CC+} = 13.2 V, V _{CC-} = -13.2 V	9	10.5		9	10.5		
V _{OL}	Low-level output voltage	V _{IH} = 1.9 V, R _L = 3 kΩ	V _{CC+} = 9 V, V _{CC-} = -9 V		-7‡	-6		-7	-6	V
			V _{CC+} = 13.2 V, V _{CC-} = -13.2 V		-10.5‡	-9		-10.5	-9	
I _{IH}	High-level input current	V _I = 5 V					10			μA
I _{IL}	Low-level input current	V _I = 0					-1 -1.6			mA
I _{OS(H)}	Short-circuit output current at high level§	V _I = 0.8 V,	V _O = 0	-4.6	-9	-13.5	-6	-9	-12	mA
I _{OS(L)}	Short-circuit output current at low level§	V _I = 1.9 V,	V _O = 0	4.6	9	13.5	6	9	12	mA
r _o	Output resistance, power off	V _{CC+} = 0, V _O = -2 V to 2 V	V _{CC-} = 0,	300			300			Ω
I _{CC+}	Supply current from V _{CC+}	V _{CC+} = 9 V, No load	All inputs at 1.9 V	15 20		15 20				mA
			All inputs at 0.8 V	4.5 6		4.5 6				
		V _{CC+} = 12 V, No load	All inputs at 1.9 V	19 25		19 25				
			All inputs at 0.8 V	5.5 7		5.5 7				
		V _{CC+} = 15 V, No load, T _A = 25°C	All inputs at 1.9 V	34		34				
			All inputs at 0.8 V	12		12				
I _{CC-}	Supply current from I _{CC-}	V _{CC-} = -9 V, No load	All inputs at 1.9 V	-13 -17		-13 -17				mA
			All inputs at 0.8 V	-0.5		-0.015				
		V _{CC-} = -12 V, No load	All inputs at 1.9 V	-18 -23		-18 -23				
			All inputs at 0.8 V	-0.5		-0.015				
		V _{CC-} = -15 V, No load, T _A = 25°C	All inputs at 1.9 V	-34		-34				
			All inputs at 0.8 V	-2.5		-2.5				
P _D	Total power dissipation	V _{CC+} = 9 V, No load		V _{CC-} = -9 V,			333			mW
		V _{CC+} = 12 V, No load		V _{CC-} = -12 V,			576			

† All typical values are at $T_A = 25^\circ\text{C}$.

‡ The algebraic convention, in which the less positive (more negative) limit is designated as minimum, is used in this data sheet for logic voltage levels only, e.g., if -6 V is a maximum, the typical value is a more negative voltage.

§ Not more than one output should be shorted at a time.



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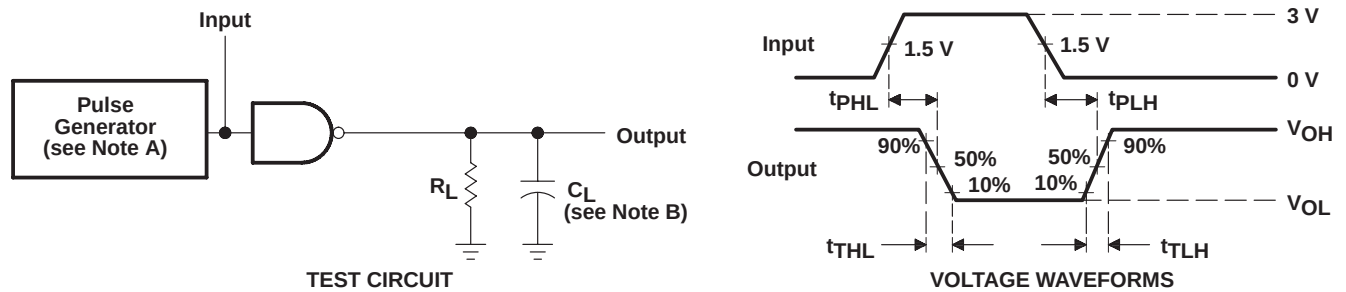
switching characteristics, $V_{CC\pm} = \pm 9\text{ V}$, $T_A = 25^\circ\text{C}$

PARAMETER	TEST CONDITIONS	MIN	TYP	MAX	UNIT
t_{PLH} Propagation delay time, low- to high-level output	$R_L = 3\text{ k}\Omega$, See Figure 1 $C_L = 15\text{ pF}$		220	350	ns
t_{PHL} Propagation delay time, high- to low-level output			100	175	ns
t_{TLH} Transition time, low- to high-level output [†]			55	100	ns
t_{THL} Transition time, high- to low-level output [†]			45	75	ns
t_{TLH} Transition time, low- to high-level output [‡]	$R_L = 3\text{ k}\Omega$ to $7\text{ k}\Omega$, See Figure 1 $C_L = 2500\text{ pF}$		2.5		μs
t_{THL} Transition time, high- to low-level output [‡]			3.0		μs

[†] Measured between 10% and 90% points of output waveform

[‡] Measured between 3 V and -3 V points on the output waveform (TIA/EIA-232-E conditions)

PARAMETER MEASUREMENT INFORMATION



- NOTES: A. The pulse generator has the following characteristics: $t_W = 0.5\text{ }\mu\text{s}$, $\text{PRR} \leq 1\text{ MHz}$, $Z_O = 50\text{ }\Omega$.
B. C_L includes probe and jig capacitance.

Figure 1. Test Circuit and Voltage Waveforms

TYPICAL CHARACTERISTICS†

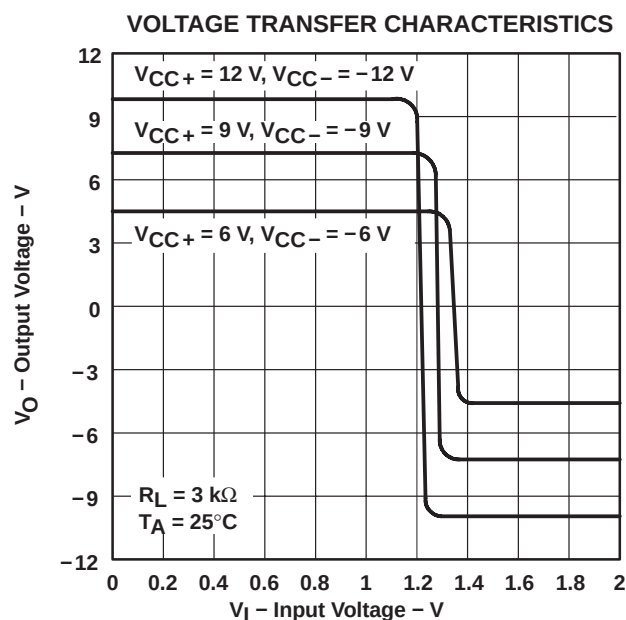


Figure 2

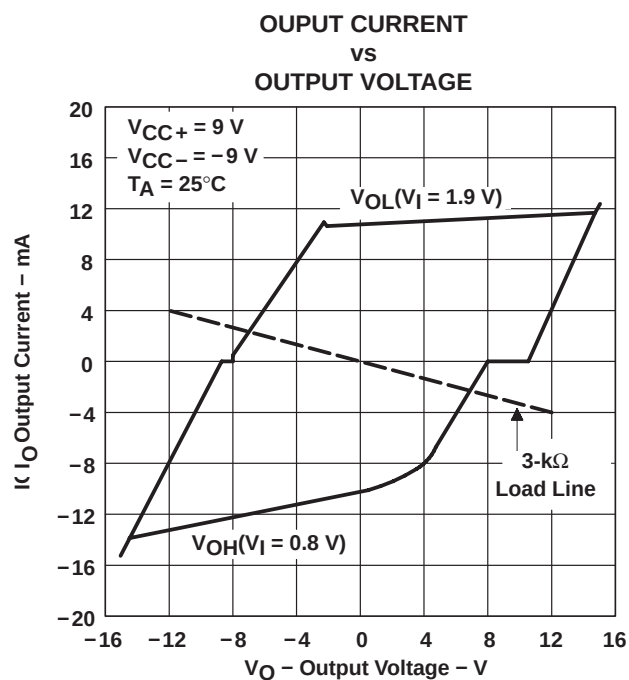


Figure 3

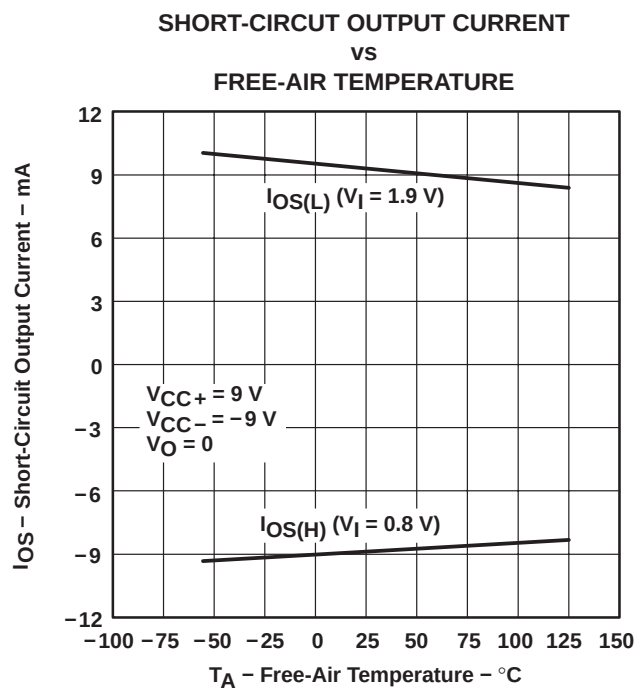


Figure 4

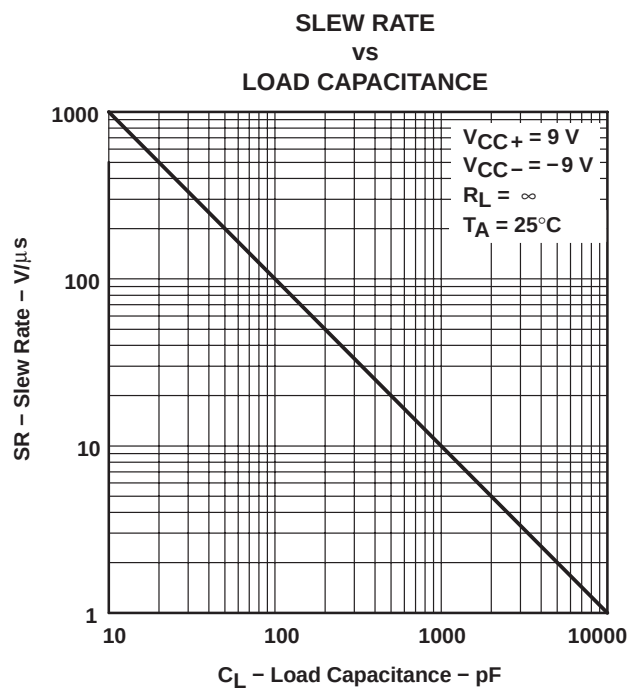


Figure 5

† Data for temperatures below 0°C and above 70°C are applicable to SN55188 circuit only.

THERMAL INFORMATION†

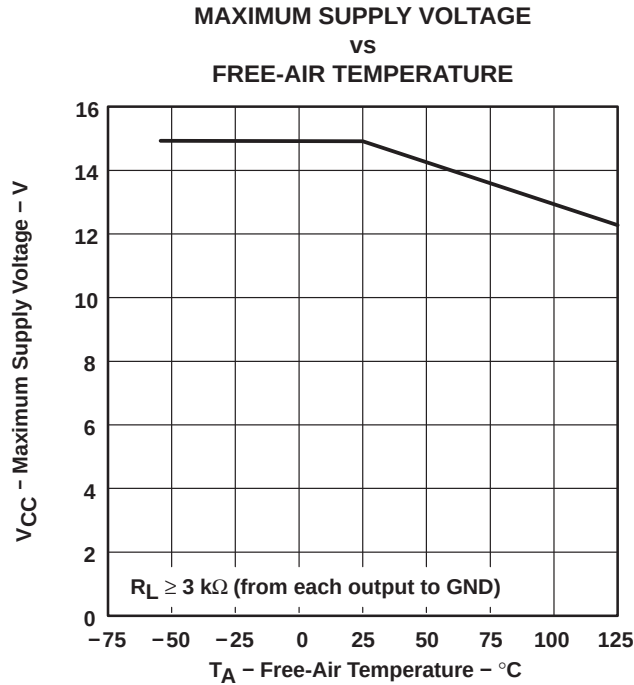


Figure 6

† Data for temperatures below 0°C and above 70°C are applicable to the SN55188 circuit only.

APPLICATION INFORMATION

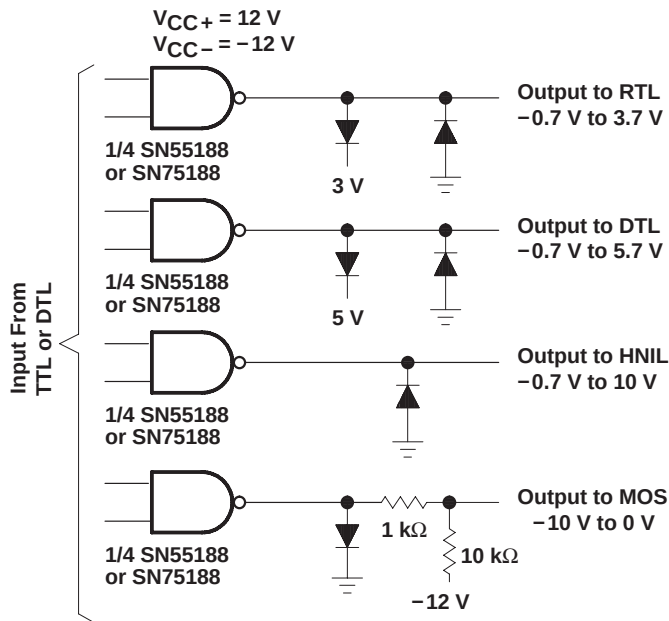
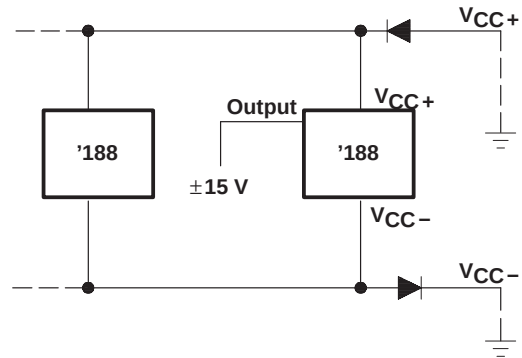


Figure 7. Logic Translator Applications



Diodes placed in series with the V_{CC+} and V_{CC-} leads protect the SN55188/SN75188 in the fault condition in which the device outputs are shorted to $\pm 15 \text{ V}$, and the power supplies are at low voltage and provide low-impedance paths to ground.

Figure 8. Power-Supply Protection to Meet
Power-Off Fault Conditions of
ANSI TIA/EIA-232-E

PACKAGING INFORMATION

Orderable part number	Status (1)	Material type (2)	Package Pins	Package qty Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
5962-86889012A	Active	Production	LCCC (FK) 20	55 TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	5962- 86889012A SNJ55 188FK
5962-8688901CA	Active	Production	CDIP (J) 14	25 TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	5962-8688901CA SNJ55188J
5962-8688901DA	Active	Production	CFP (W) 14	25 TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	5962-8688901DA SNJ55188W
MC1488N	Obsolete	Production	PDIP (N) 14	-	-	Call TI	Call TI	0 to 70	MC1488N
SN55188J	Active	Production	CDIP (J) 14	25 TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	SN55188J
SN55188J.A	Active	Production	CDIP (J) 14	25 TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	SN55188J
SN75188D	Obsolete	Production	SOIC (D) 14	-	-	Call TI	Call TI	0 to 70	SN75188
SN75188DR	Active	Production	SOIC (D) 14	2500 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	0 to 70	SN75188
SN75188DR.A	Active	Production	SOIC (D) 14	2500 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	0 to 70	SN75188
SN75188N	Active	Production	PDIP (N) 14	25 TUBE	Yes	NIPDAU	N/A for Pkg Type	0 to 70	SN75188N
SN75188N.A	Active	Production	PDIP (N) 14	25 TUBE	Yes	NIPDAU	N/A for Pkg Type	0 to 70	SN75188N
SN75188NSR	Active	Production	SOP (NS) 14	2000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	0 to 70	SN75188
SN75188NSR.A	Active	Production	SOP (NS) 14	2000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	0 to 70	SN75188
SNJ55188FK	Active	Production	LCCC (FK) 20	55 TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	5962- 86889012A SNJ55 188FK
SNJ55188FK.A	Active	Production	LCCC (FK) 20	55 TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	5962- 86889012A SNJ55 188FK
SNJ55188J	Active	Production	CDIP (J) 14	25 TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	5962-8688901CA SNJ55188J
SNJ55188J.A	Active	Production	CDIP (J) 14	25 TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	5962-8688901CA SNJ55188J
SNJ55188W	Active	Production	CFP (W) 14	25 TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	5962-8688901DA SNJ55188W

Orderable part number	Status (1)	Material type (2)	Package Pins	Package qty Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
SNJ55188W.A	Active	Production	CFP (W) 14	25 TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	5962-8688901DA SNJ55188W

- ⁽¹⁾ **Status:** For more details on status, see our [product life cycle](#).
- ⁽²⁾ **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.
- ⁽³⁾ **RoHS values:** Yes, No, RoHS Exempt. See the [TI RoHS Statement](#) for additional information and value definition.
- ⁽⁴⁾ **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.
- ⁽⁵⁾ **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.
- ⁽⁶⁾ **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

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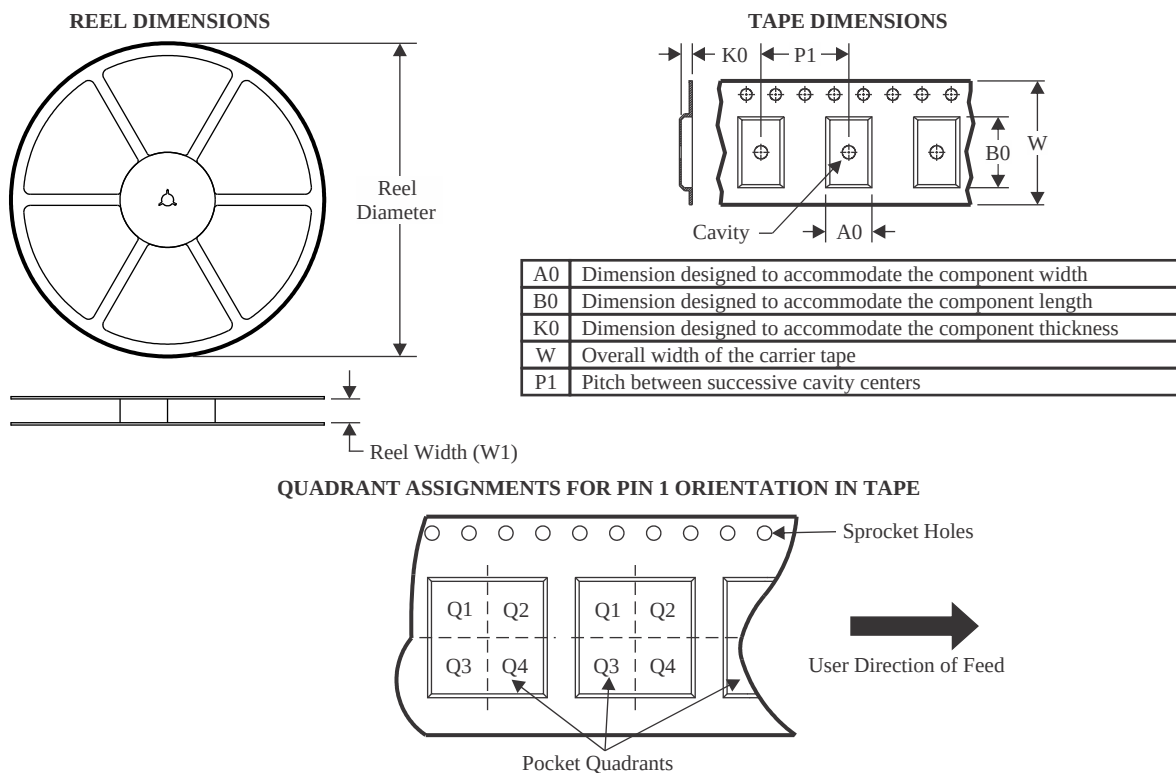
OTHER QUALIFIED VERSIONS OF SN55188, SN75188 :

- Catalog : [SN75188](#)
- Military : [SN55188](#)

NOTE: Qualified Version Definitions:

- Catalog - TI's standard catalog product
- Military - QML certified for Military and Defense Applications

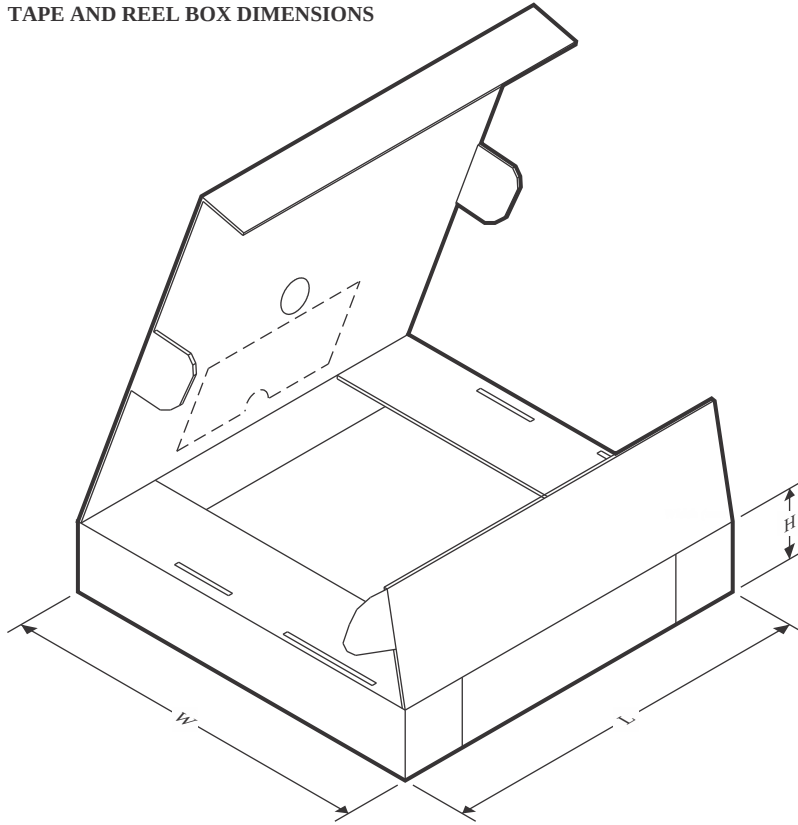
TAPE AND REEL INFORMATION



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
SN75188DR	SOIC	D	14	2500	330.0	16.4	6.5	9.0	2.1	8.0	16.0	Q1
SN75188NSR	SOP	NS	14	2000	330.0	16.4	8.2	10.5	2.5	12.0	16.0	Q1

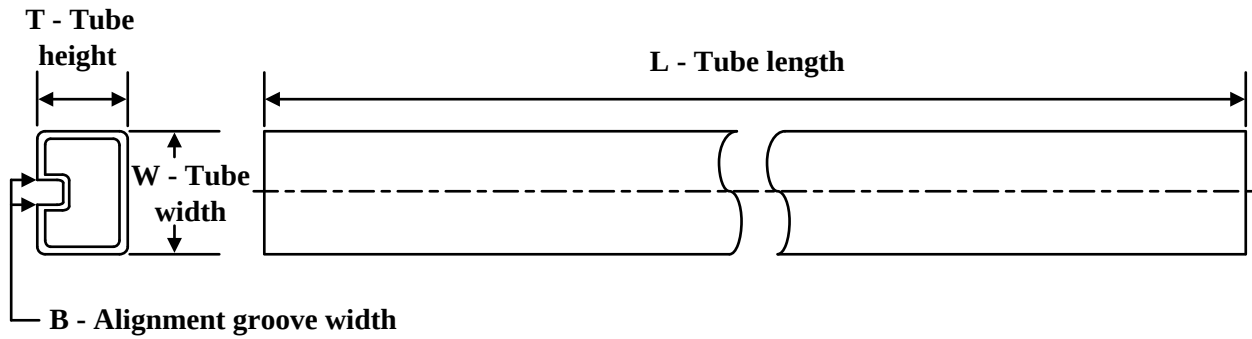
TAPE AND REEL BOX DIMENSIONS



*All dimensions are nominal

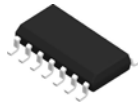
Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
SN75188DR	SOIC	D	14	2500	356.0	356.0	35.0
SN75188NSR	SOP	NS	14	2000	356.0	356.0	35.0

TUBE

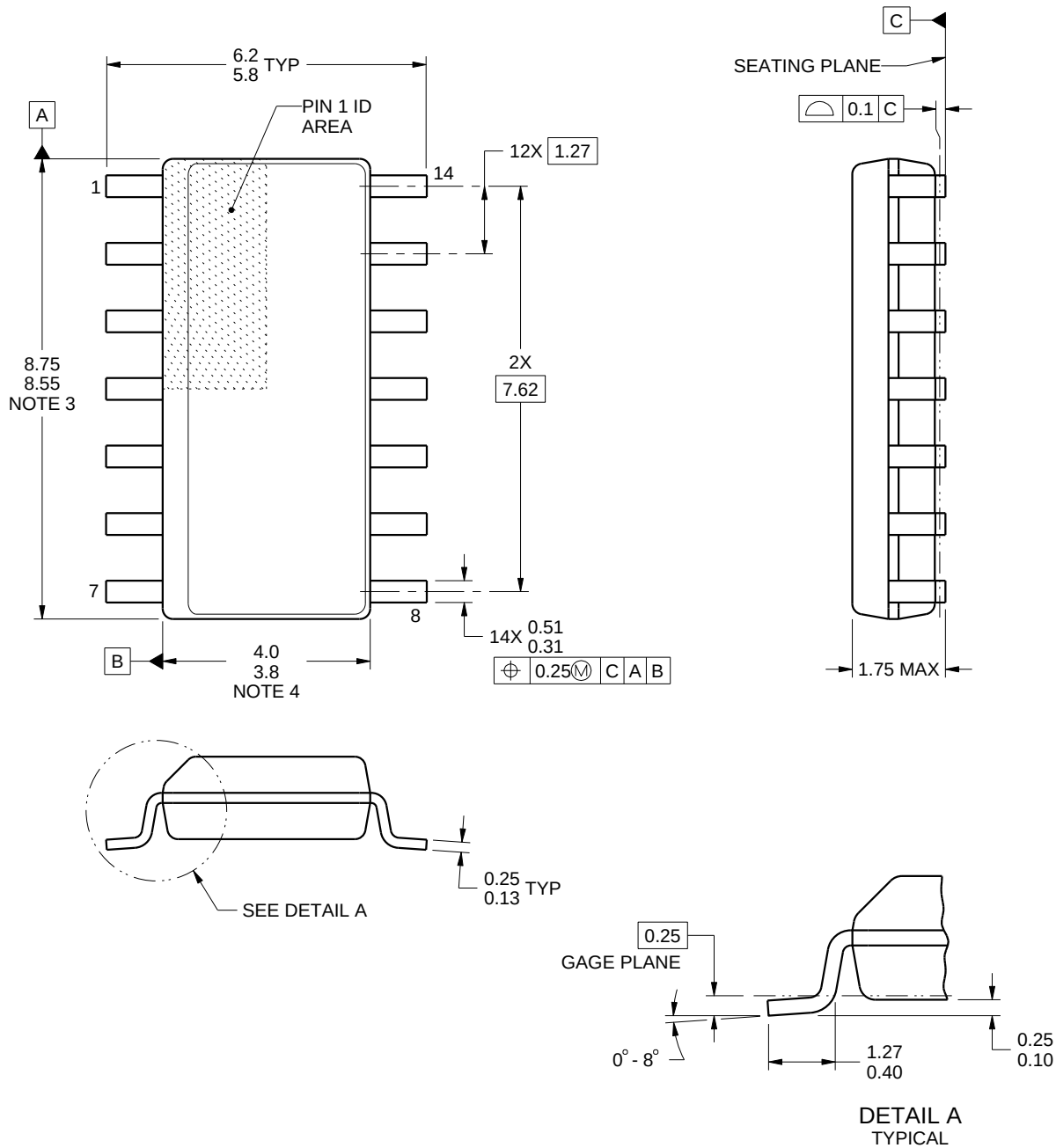


*All dimensions are nominal

Device	Package Name	Package Type	Pins	SPQ	L (mm)	W (mm)	T (μm)	B (mm)
5962-86889012A	FK	LCCC	20	55	506.98	12.06	2030	NA
5962-8688901DA	W	CFP	14	25	506.98	26.16	6220	NA
SN75188N	N	PDIP	14	25	506	13.97	11230	4.32
SN75188N	N	PDIP	14	25	506	13.97	11230	4.32
SN75188N.A	N	PDIP	14	25	506	13.97	11230	4.32
SN75188N.A	N	PDIP	14	25	506	13.97	11230	4.32
SNJ55188FK	FK	LCCC	20	55	506.98	12.06	2030	NA
SNJ55188FK.A	FK	LCCC	20	55	506.98	12.06	2030	NA
SNJ55188W	W	CFP	14	25	506.98	26.16	6220	NA
SNJ55188W.A	W	CFP	14	25	506.98	26.16	6220	NA

D0014A**PACKAGE OUTLINE****SOIC - 1.75 mm max height**

SMALL OUTLINE INTEGRATED CIRCUIT



4220718/A 09/2016

NOTES:

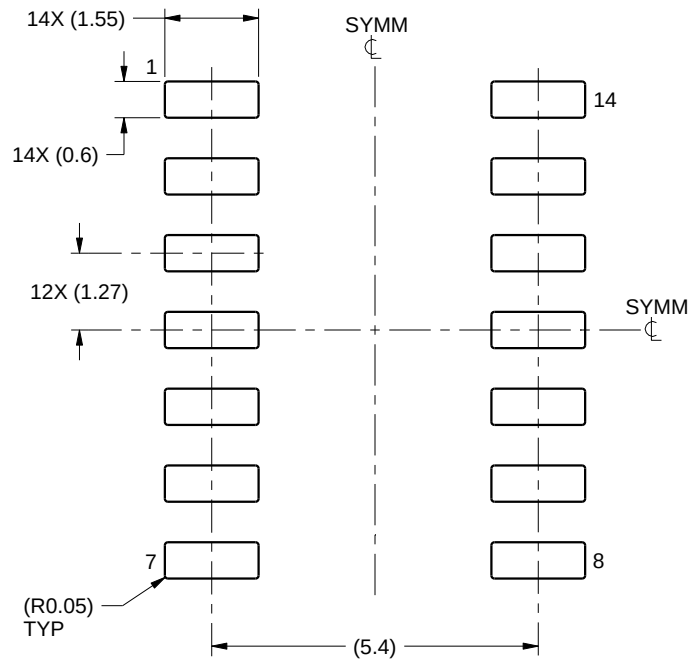
1. All linear dimensions are in millimeters. Dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm, per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.43 mm, per side.
5. Reference JEDEC registration MS-012, variation AB.

EXAMPLE BOARD LAYOUT

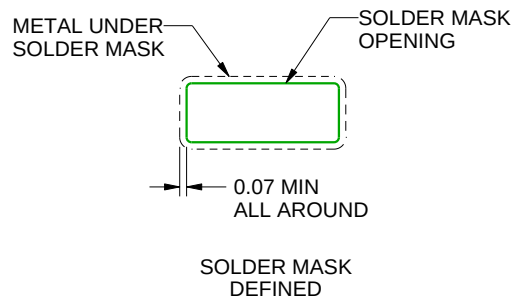
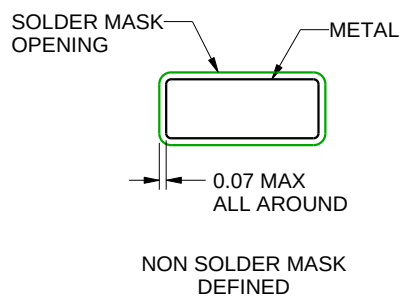
D0014A

SOIC - 1.75 mm max height

SMALL OUTLINE INTEGRATED CIRCUIT



LAND PATTERN EXAMPLE
SCALE:8X



SOLDER MASK DETAILS

4220718/A 09/2016

NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.

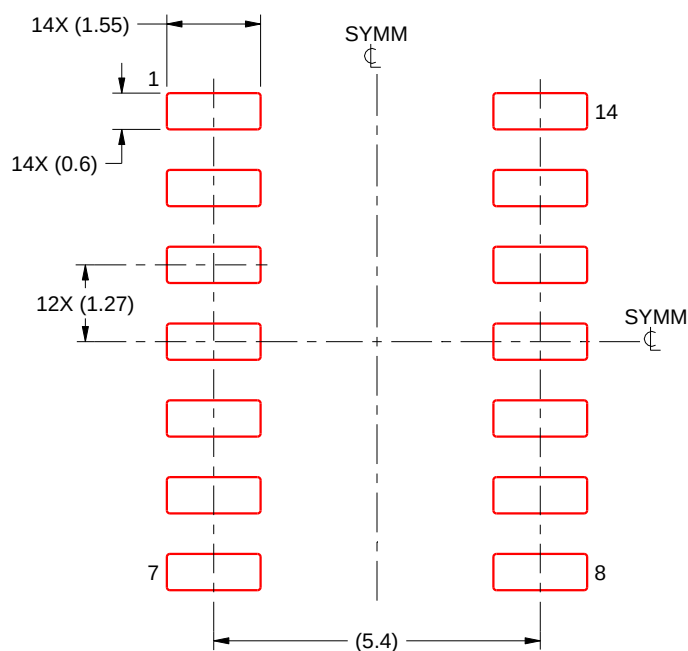
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

EXAMPLE STENCIL DESIGN

D0014A

SOIC - 1.75 mm max height

SMALL OUTLINE INTEGRATED CIRCUIT



SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL
SCALE:8X

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NOTES: (continued)

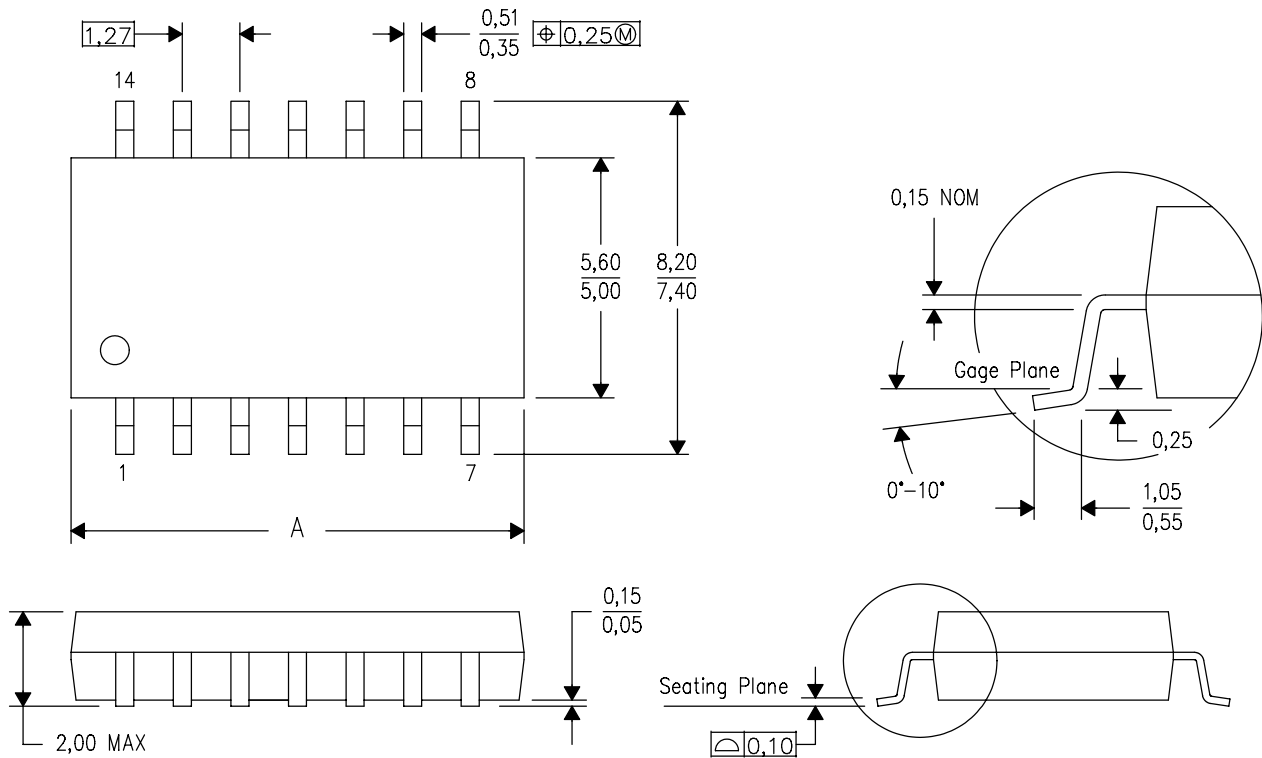
8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

MECHANICAL DATA

NS (R-PDSO-G**)

PLASTIC SMALL-OUTLINE PACKAGE

14-PINS SHOWN



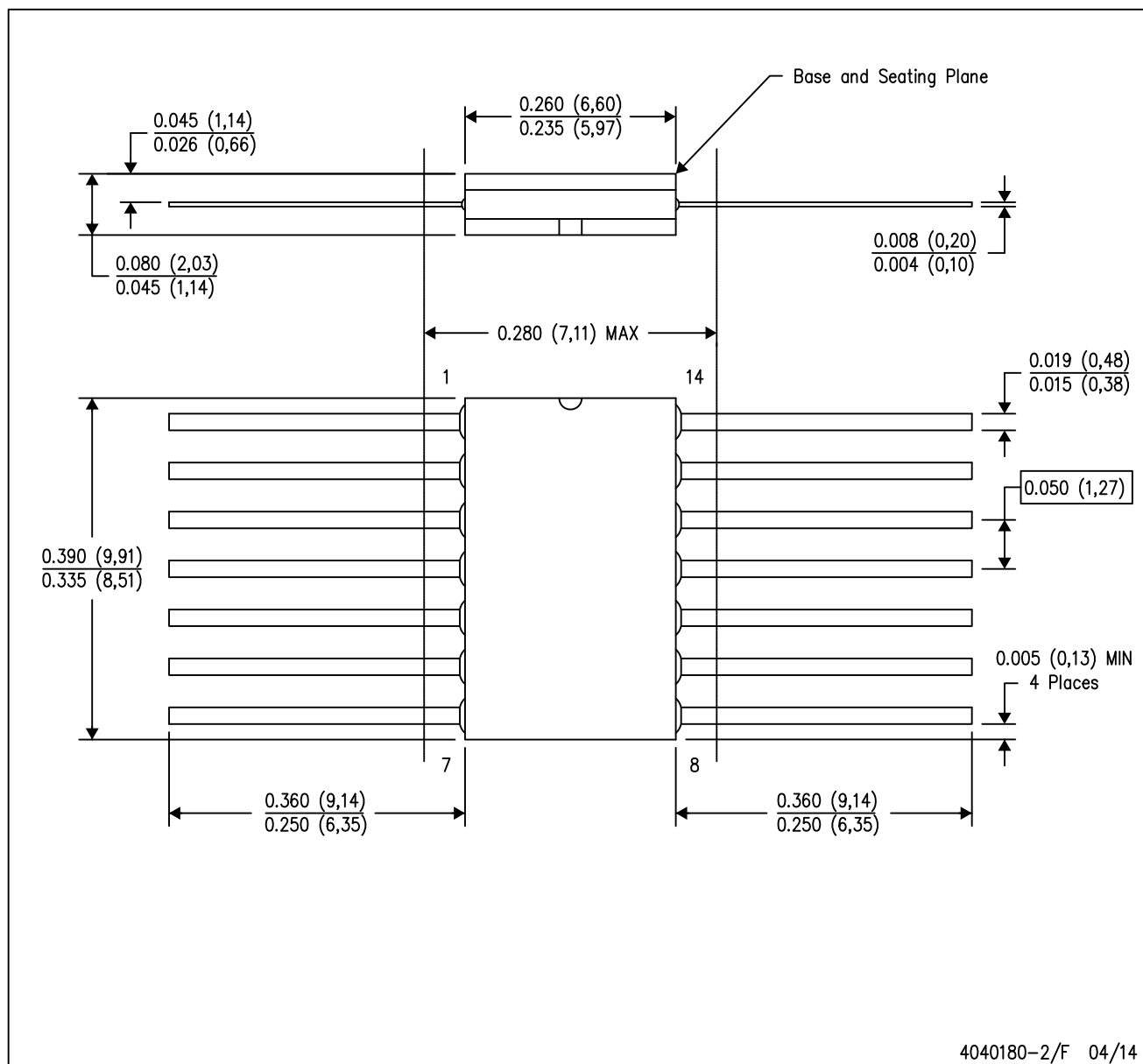
DIM \ PINS **	14	16	20	24
A MAX	10,50	10,50	12,90	15,30
A MIN	9,90	9,90	12,30	14,70

4040062/C 03/03

- NOTES:
- A. All linear dimensions are in millimeters.
 - B. This drawing is subject to change without notice.
 - C. Body dimensions do not include mold flash or protrusion, not to exceed 0,15.

W (R-GDFP-F14)

CERAMIC DUAL FLATPACK



GENERIC PACKAGE VIEW

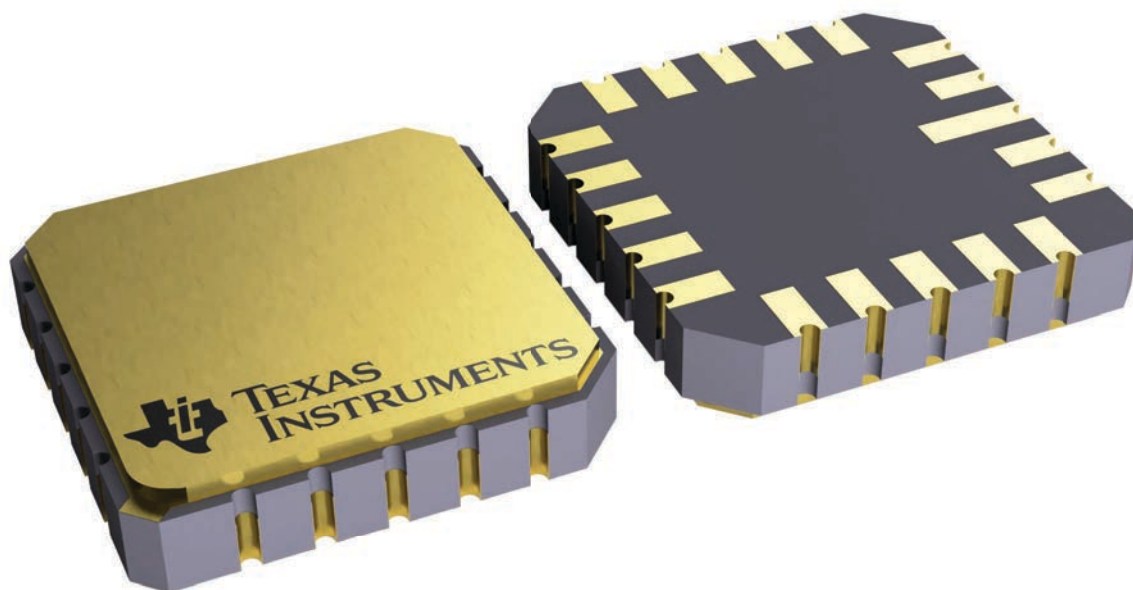
FK 20

LCCC - 2.03 mm max height

8.89 x 8.89, 1.27 mm pitch

LEADLESS CERAMIC CHIP CARRIER

This image is a representation of the package family, actual package may vary.
Refer to the product data sheet for package details.



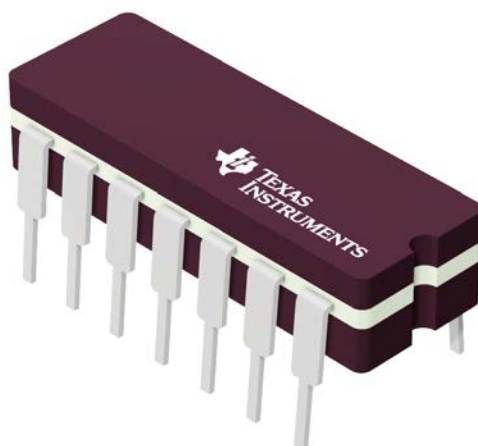
4229370VA\

J 14

GENERIC PACKAGE VIEW

CDIP - 5.08 mm max height

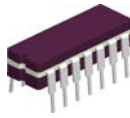
CERAMIC DUAL IN LINE PACKAGE



Images above are just a representation of the package family, actual package may vary.
Refer to the product data sheet for package details.

4040083-5/G

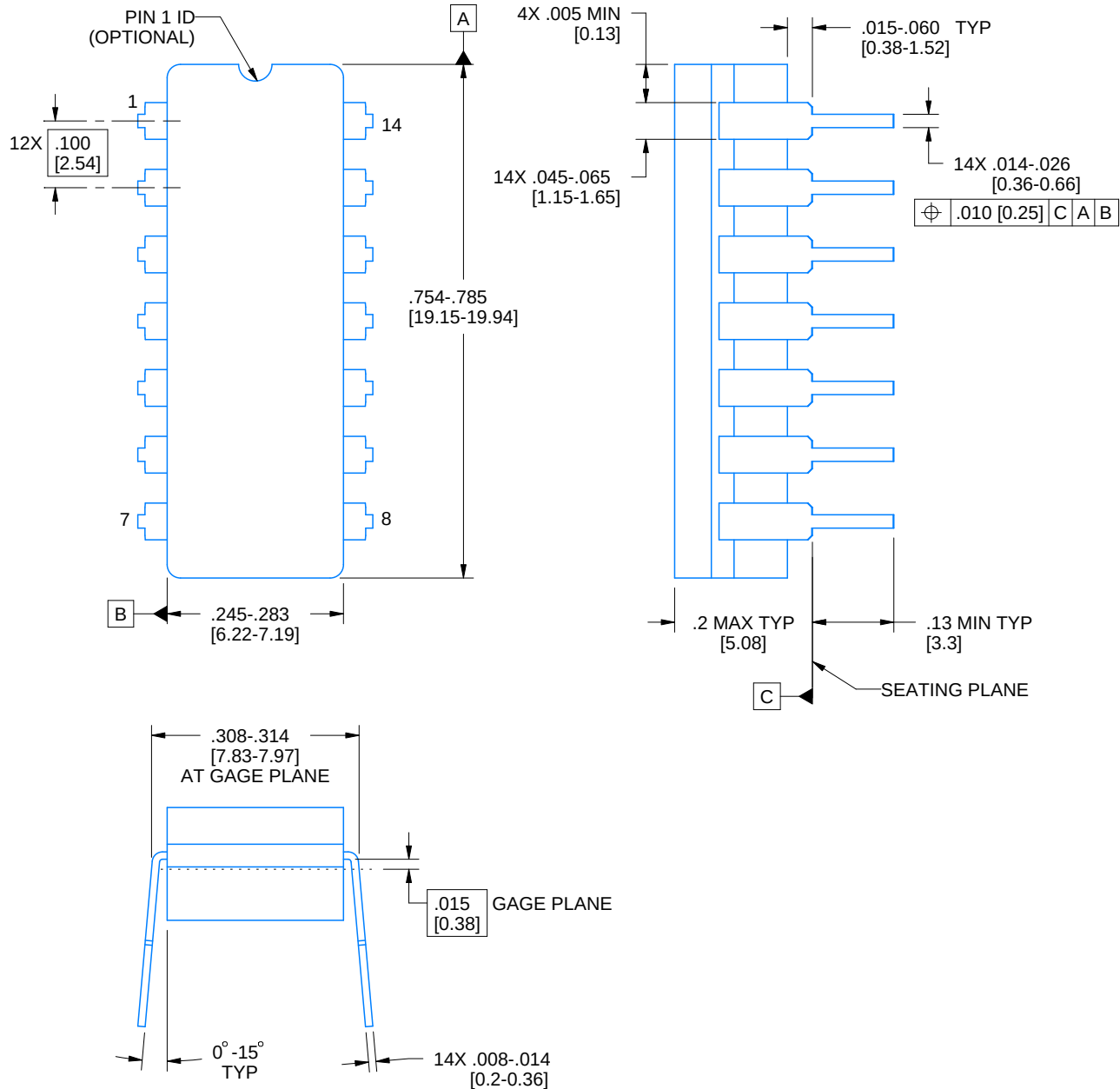
J0014A



PACKAGE OUTLINE

CDIP - 5.08 mm max height

CERAMIC DUAL IN LINE PACKAGE



4214771/A 05/2017

NOTES:

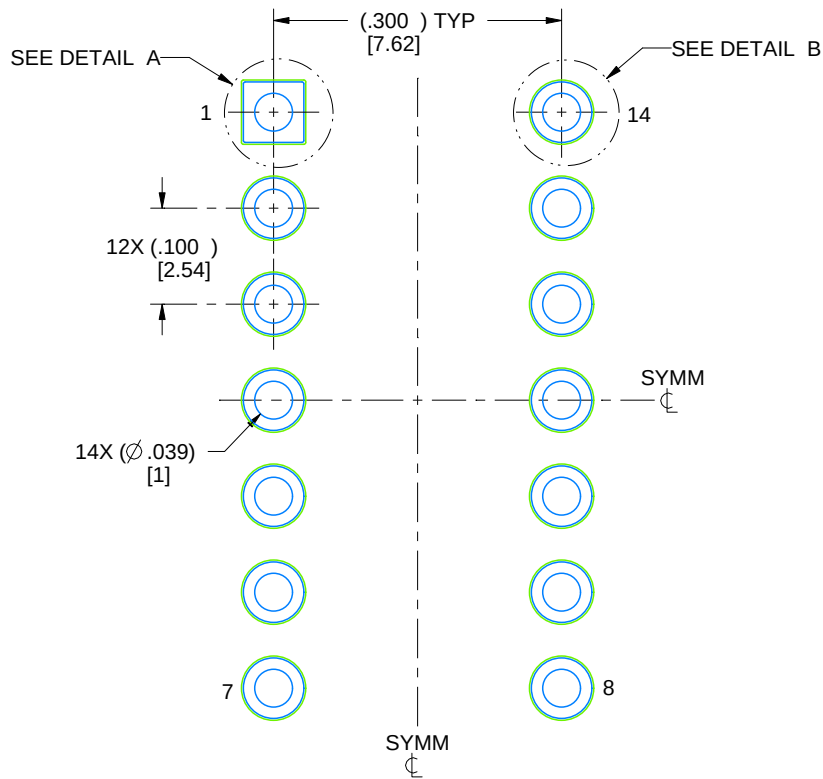
1. All controlling linear dimensions are in inches. Dimensions in brackets are in millimeters. Any dimension in brackets or parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This package is hermetically sealed with a ceramic lid using glass frit.
4. Index point is provided on cap for terminal identification only and on press ceramic glass frit seal only.
5. Falls within MIL-STD-1835 and GDIP1-T14.

EXAMPLE BOARD LAYOUT

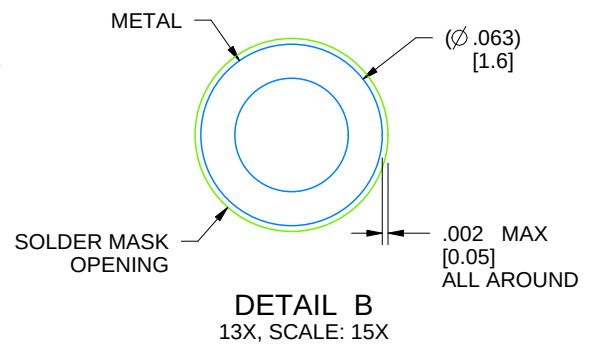
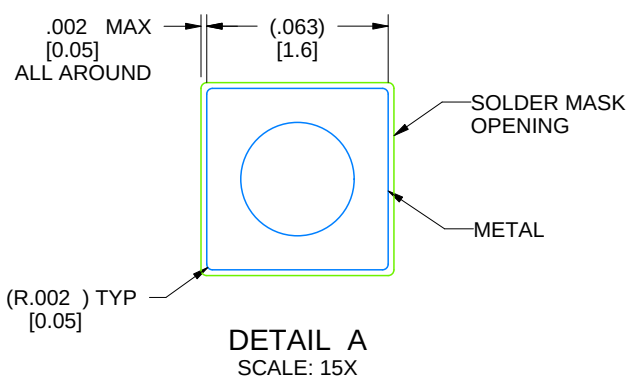
J0014A

CDIP - 5.08 mm max height

CERAMIC DUAL IN LINE PACKAGE



LAND PATTERN EXAMPLE
NON-SOLDER MASK DEFINED
SCALE: 5X

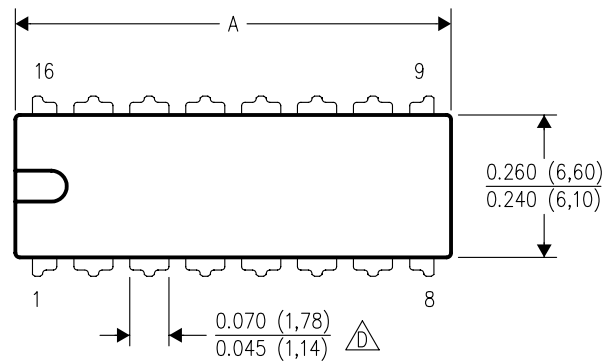


4214771/A 05/2017

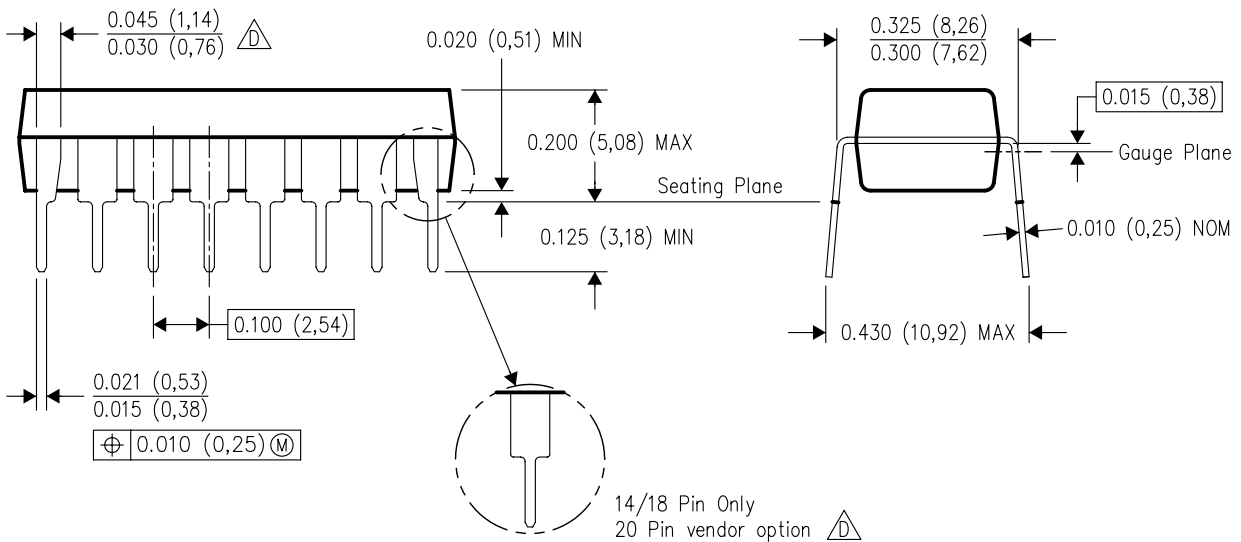
N (R-PDIP-T**)

16 PINS SHOWN

PLASTIC DUAL-IN-LINE PACKAGE



PINS **	14	16	18	20
DIM				
A MAX	0.775 (19,69)	0.775 (19,69)	0.920 (23,37)	1.060 (26,92)
A MIN	0.745 (18,92)	0.745 (18,92)	0.850 (21,59)	0.940 (23,88)
MS-001 VARIATION	AA	BB	AC	AD



4040049/E 12/2002

- NOTES:
- A. All linear dimensions are in inches (millimeters).
 - B. This drawing is subject to change without notice.
 - Falls within JEDEC MS-001, except 18 and 20 pin minimum body length (Dim A).
 - The 20 pin end lead shoulder width is a vendor option, either half or full width.

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